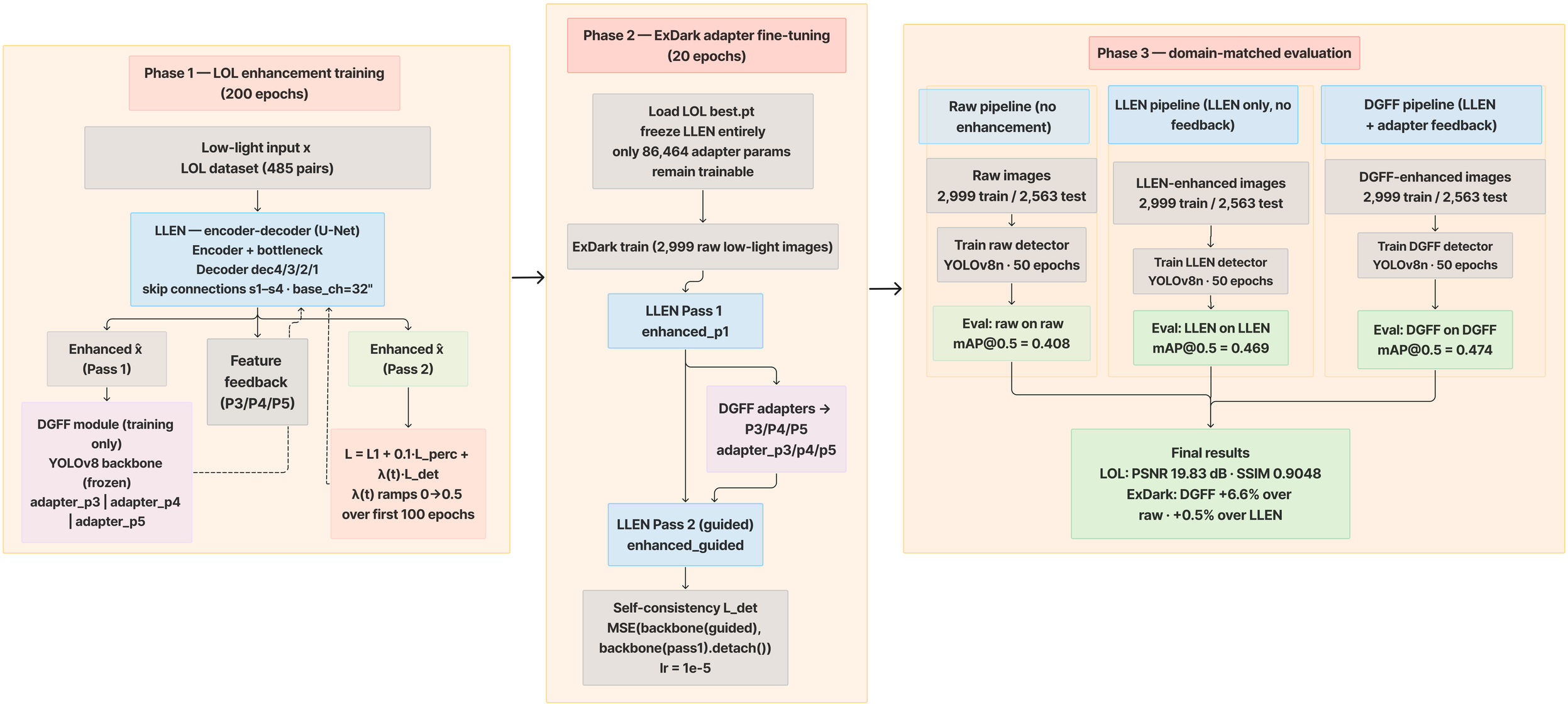
Phase 1 — LOL enhancement training (200 epochs)
Low-light input x
LOL dataset (485 pairs)
LLEN — encoder-decoder (U-Net)
Encoder + bottleneck
Decoder dec4/3/2/1
skip connections s1–s4 · base_ch=32"
Enhanced x̂ (Pass 1)
Feature feedback (P3/P4/P5)
Enhanced x̂ (Pass 2)
DGFF module (training only)
YOLOv8 backbone (frozen)
adapter_p3 | adapter_p4 | adapter_p5
L = L1 + 0.1·L_perc + λ(t)·L_det
λ(t) ramps 0→0.5 over first 100 epochs
Phase 2 — ExDark adapter fine-tuning (20 epochs)
Load LOL best.pt
freeze LLEN entirely
only 86,464 adapter params remain trainable
ExDark train (2,999 raw low-light images)
LLEN Pass 1
enhanced_p1
DGFF adapters → P3/P4/P5
adapter_p3/p4/p5
LLEN Pass 2 (guided)
enhanced_guided
Self-consistency L_det
MSE(backbone(guided), backbone(pass1).detach())
lr = 1e-5
Phase 3 — domain-matched evaluation
Raw pipeline (no enhancement)
LLEN pipeline (LLEN only, no feedback)
DGFF pipeline (LLEN + adapter feedback)
Raw images
2,999 train / 2,563 test
LLEN-enhanced images
2,999 train / 2,563 test
DGFF-enhanced images
2,999 train / 2,563 test
Train raw detector
YOLOv8n · 50 epochs
Train LLEN detector
YOLOv8n · 50 epochs
Train DGFF detector
YOLOv8n · 50 epochs
Eval: raw on raw
mAP@0.5 = 0.408
Eval: LLEN on LLEN
mAP@0.5 = 0.469
Eval: DGFF on DGFF
mAP@0.5 = 0.474
Final results
LOL: PSNR 19.83 dB · SSIM 0.9048
ExDark: DGFF +6.6% over raw · +0.5% over LLEN